Khulna University of Engineering & Technology
B.Sc. Engineering 2nd Year 1st Term Examination, 2020
Department of Materials Science and Engineering

MSE 2101
(Crystallography)

Time: 1.5 hours

Full Marks: 120

N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY TWO** questions from this section in **answer script A**)

Q 1. (a) Define primitive and non-primitive unit cell. Describe different types of atomic arrangement with figures. (08)

(b) What is space and lattice and basis? Determine the relationship between atomic radius (r) and lattice volume (V) in BCC and calculate atomic packing factor (APF). (12)

(c) Define Wigner-Seitz cell? Draw and explain the Wigner-Seitz cell on the following space lattices (Figure 1). (10)

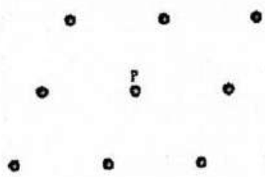

Figure 1: Lattice points.

Q 2. (a) Define mirror and rotational symmetry. Determine the Miller indices for the planes shown in the following two cubic unit cells (Figure 2). (10)

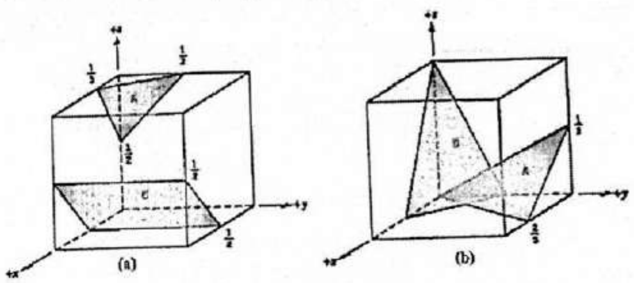

Figure 2: Planes in the cubic unit cell.

(b) What is Bravais lattice? Explain any two missing Bravais lattices with figures. (08)

(c) What is planar packing density? Draw the following planes and directions in cubic unit cell. (12)

$(1\bar{1}\bar{1}), (\bar{3}3\bar{1}), (\bar{2}1\bar{2})$ and $[\bar{3}31], [1\bar{2}\bar{2}], [321]$

Q 3. (a) Define and draw distorted tetrahedral and octahedral voids? For distorted tetrahedral voids shown that $r = 0.29R$ in BCC. [Where r = Radius of tetrahedral voids, R = Radius of atoms] (10)

(b) What is symmetry operation? Draw (011) and (111) planes in a cubic unit cell determine the Miller indicies of direction which are common in both planes. (08)

(c) Give the relationship between inter-planar spacing (d_{hkl}) and Miller indices (hkl) for tetrahedral system. (12)

Section B

(Answer **ANY TWO** questions from this section in **answer script B**)

Q 4. (a) Define reciprocal lattice. Show that inter-planar spacing of (110) plane in reciprocal lattice is reciprocal to the corresponding inter-planar spacing in normal lattice. (10)

(b) What is zone axis? Derive the Weiss zone law for n^{th} parallel lattice plane. (10)

(c) Draw a side view of a monoclinic crystal with Miller indices perpendicular to the Y-axis. Now construct the pattern of reciprocal lattice points from this side view of the crystal. (10)

Q 5. (a) What do you mean by X-ray diffraction? Differentiate between X-ray diffraction and reflection of visible light. (10)

(b) For BCC iron, the lattice parameter of iron is 0.2866 nm. Suppose a monochromatic radiation having wave length of 0.179 nm is used and the order of reflection is 1. Then calculate: (10)

(i) The inter-planar spacing

(ii)The diffraction angle for the (220) set of planes.

(c) What do you mean by forbidden reflection? "Bragg's law is a necessary but not sufficient condition for diffraction by real crystals". Justify with your answer. (10)

Q 6. (a) Write down the experimental conditions for different diffraction methods. Briefly discuss the Debye-Scherrer method for diffraction. (10)

(b) A X-ray beam have a minimum wavelength of 0.5 Å and makes a 60° angle with the surface of the crystal [d_{100} = 1 Å]. So, how many orders of reflections are possible and why? (10)

(c) Draw a stereographic projection of a cubic crystal on (110), $(0\bar{1}0)$, (001) planes. If the cubic crystal plane (111) shares a zone with plane (110) and (001). Then plot (111) plane on this stereographic projection and calculate the angle between 110 and 111 stereographic poles. (10)

Khulna University of Engineering & Technology
B.Sc. Engineering 2nd Year 1st Term Examination, 2020
Department of Materials Science and Engineering

MSE 2103
(Thermodynamics of Materials)

Time: 1.5 hours Full Marks: 120

N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks
(iii) Assume reasonable data if any missing

Section A

(Answer **ANY TWO** questions from this section in **answer script A**)

Q 1. (a) Distinguish between the followings: (20)

i) State variables and process variables
ii) Intensive properties and extensive properties
iii) Path and process
iv) Adiabatic process and isochoric process
v) Stable equilibrium and metastable equilibrium.

(b) The functional relation between three variables x, y, z can be represented as $z = 4x^2y + 3xy^3$. Determine the total differential equation of the function $z = z(x, y)$ and then obtain the coefficient relations. Are these variables state functions? (10)

Q 2. (a) "The first law of thermodynamics deals with the conservation of energy while the second law deals with the degradation of energy"-Explain. (08)

(b) How can you quantify irreversibility? (11)

(c) One mole of an ideal gas is compressed in a piston-cylinder device from 2 bar and 25°C to 7 bar. The process is irreversible and requires 35% more work than a reversible adiabatic compression from the same initial state to the same final pressure. What is the entropy change of the gas? (11)

Q 3. (a) Deduce a relationship for the change in internal energy of the system to its pressure and volume. (12)

(b) 5 moles of an ideal diatomic gas, initially at 15 atm pressure and 27°C temperature, is expanded thrice of its volume by a reversible isothermal process. Calculate the change in internal energy of the system. (10)

(c) State the equilibrium criterion for internal energy, enthalpy, Helmholtz free energy and Gibbs free energy. (08)

Section B

(Answer **ANY TWO** questions from this section in **answer script B**)

Q 4. (a) Derive the Clasius-Clapeyron equation which is used most frequently in computing unary phase diagrams. (15)

(b) Construct the phase diagram for pure silicon using the following information: $T_m = 1683\ K$, $T_b = 2750\ K, \Delta H^m = 46.4\ kJ/mol, \Delta H^v = 297\ kJ/mol.$ (15)

Q 5. (a) Define partial molar properties. Explain the consequences of the definition of partial molar properties. (15)

(b) Define fugacity and activity coefficient. Show that for any component in an ideal gas mixture, activity is equal to mole fraction and activity coefficient is equal to one. (15)

Q 6. (a) Define variance of a reacting system. Deduce a relation showing the dependence of equilibrium constant for any chemical reaction on temperature. (10)

(b) Explain the prominent features of Ellingham diagram with related examples. (20)

Khulna University of Engineering & Technology
B.Sc. Engineering 2nd Year 1st Term Examination, 2020
Department of Materials Science and Engineering

Math 2127
(Vector Analysis and Linear Algebra)

Time: 1.5 hours — Full Marks: 120

N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks
(iii) Assume reasonable data if any missing

Section A

(Answer **ANY TWO** questions from this section in **answer script A**)

Q 1. (a) Discuss with example about the arithmetic operations and rules in matrix which are different from that operations and rules in real numbers. (12)

(b) Reduce the following matrix to its echelon form, canonical form and finally normal form sequentially. $\begin{bmatrix} 0 & 1 & p & -2 \\ 1 & 1 & q & 1 \\ 2 & 1 & 1 & 0 \end{bmatrix}$. Hence find its Rank. Where p and q are the last two digits of your class roll number measured from the rightmost sequentially. (See Hints) (18)

Q 2. (a) Find the inverse of A, if possible, by using elementary transformation, where (18)

$$\mathbf{A}=\begin{bmatrix} 1 & 2 & -1 \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{bmatrix}$$

Now given that AX= H, where, H = $[1 \quad q \quad p]^t$. Where p and q are the last two digits of your class roll number measured from the rightmost sequentially. (See Hints) Then find the solution of X.

(b) Find the conditions on λ and μ so that the following system of linear equations will have (i) a unique solution, (ii) no solution and (iii) many solutions. (12)

$$2x + 3y + z = 5; \qquad 3x - y + \lambda z = 2; \qquad x + 7y - 6z = \mu$$

Q 3. (a) Find the Eigen values and Eigen vector corresponding to the largest Eigen value of A, if exist. (15)
$\mathbf{A}=\begin{pmatrix} 2 & q+1 \\ p+1 & 1 \end{pmatrix}$, where p and q are the last two digits of your class roll number measured from the rightmost sequentially. (See Hints)

(b) Let X= $[1 \quad q \quad p]^t$, Y= $[1 \quad 2 \quad 3]^t$, Z= $[1 \quad 1 \quad 0]^t$ are they formed vector space $\mathbb{R}^3$ (i.e. are they linearly independent vector of Euclidean vector space $\mathbb{R}^3$)? If not, then express the dependent vector as a linear combination of independent vectors in $\mathbb{R}^2$. Where p and q are the last two digits of your class roll number measured from the rightmost sequentially. (See Hints) (10)

(c) Define Row space and Null space. (05)

[Hints: Roll 1827001, then p =1, q =0 and
Roll 1827023, then p = 3, q = 2]

Section B

(Answer **ANY TWO** questions from this section in **answer script B**)

Q 4. (a) Define gradient of a scalar function $\psi(x,y,z)$. Find the values of the constants a, b, and c so that the directional derivative of $\phi = axy^2 + byz + cz^2x^3$ at (1,2,-1) has a maximum of magnitude 64 in a direction parallel to the line $\frac{x-1}{2} = \frac{y-3}{-2} = \frac{z}{1}$. (15)

(b) Test the nature of the vector point function $\frac{G\,\vec{r}}{r^2}$ (by using vector operator '∇'), where G is a constant and $\vec{r}$ is the position vector. Hence find the scalar potential function ϕ such that $\phi(R,0,R) = 1$, where R is the sum of the last two digits of your class roll number. (See Hints) (15)

[Hints: Roll 1827005, then R=0+5=5 and
Roll 1827029, then R=2+9=11]

Q 5. (a) Verify Stoke's Theorem for the function $\vec{F} = x^2\hat{\imath} - xy\hat{\jmath}$ integrated round the square in the plane $z = 0$ and bounded by the lines $x = 0,\ y = 0,\ x = 2,\ y = 2$. (20)

(b) Compute $\oint (xy - x^2)dx + x^2y\,dy$ over the triangle bounded by the lines $y = 0,\ x = 1,$ $y = x$. (10)

Q 6. (a) Apply Divergence theorem to evaluate $\iint_S \vec{F}.\hat{n}\ dS$ where $\vec{F} = 4x^3\hat{\imath} - x^2y\hat{\jmath} + x^2z\hat{k}$ and S is the surface of the cylinder $x^2 + y^2 = 4$ bounded by the planes $z = 0$ and $z = 1$. (15)

(b) Derive Grad f, where f is a scalar function, in orthogonal curvilinear system. Hence derive it in cylindrical coordinates. (15)

Khulna University of Engineering & Technology
B.Sc. Engineering 2nd Year 1st Term Examination, 2020
Department of Materials Science and Engineering

ME 2127
(Solid Mechanics)

Time: 1.5 hours | Full Marks: 120

N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks
(iii) Assume reasonable data if any missing

Section A

(Answer **ANY TWO** questions from this section in **answer script A**)

Q 1. (a) Compare and contrast between truss and frame. (10)

(b) Draw the freebody diagram of the frame shown in following figure 1(b). (10)

(c) Determine the forces in the members BC, EC and JC of the truss shown in figure 1(c). (10)

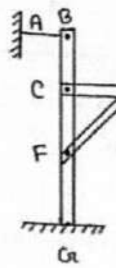

Figure: 1(b)

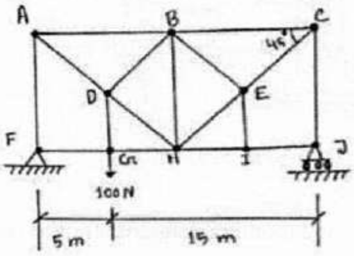

Figure: 1(c)

Q 2. (a) Under what circumstances centroid and center of gravity passes through the same point? Discuss. (05)

(b) Calculate the centroid of the shaded area by direct integration in figure 2(b). (15)

(c) Calculate the abscissa of the centroid of the trapezoid shown in figure 2(c) in terms of h_1, h_2 and a. (10)

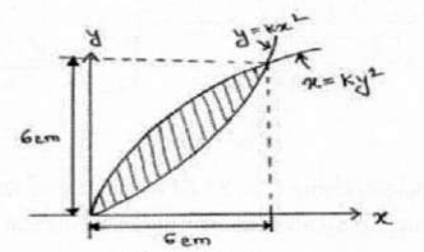

Figure: 2(b)

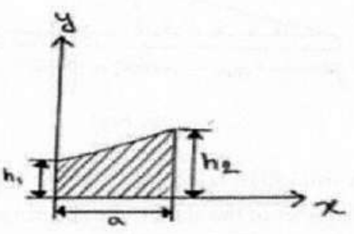

Figure: 2(c)

Q 3. (a) Determine the moment of inertia and radius of the shaded area as shown in figure 3(a) below with respect to X-axis. (15)

(b) Determine the moment of inertia of the shaded area as shown in figure 3(b) below with respect to Y-axis by direct integration. (15)

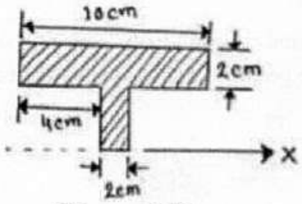

Figure: 3(a)

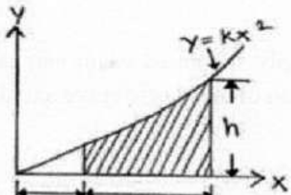

Figure: 3(b)

Section B

(Answer **ANY TWO** questions from this section in **answer script B**)

Q 4. (a) Compute the shearing stress in the pin at D for the member supported as shown in figure 4(a) below. The pin diameter is $18mm$. (12)

(b) A rigid block of weight W is supported by the symmetrically spaced rod as shown in figure 4(b) below. Each aluminum rod has an area of $800mm^2$, $E = 70GPa$ and allowable stress is $70MPa$. The copper rod has an area of $1100mm^2$, $E = 120GPa$ and the allowable stress is $140MPa$. Determine the largest weight W which can be supported. (18)

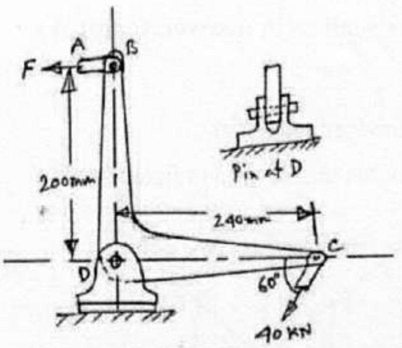

Figure: 4(a)

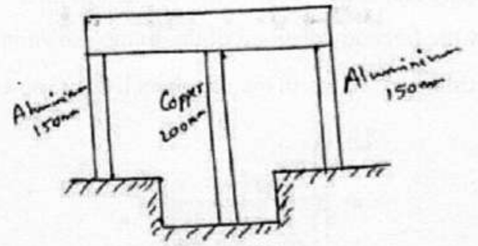

Figure: 4(b)

Q 5. (a) Draw shear and moment diagrams for the cantilever beam as shown in figure 5(a) below. (15)

(b) Determine the minimum width b of the beam as shown in figure 5(b) below if the flexural stress is not to exceed $8MPa$. (15)

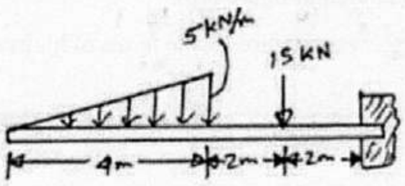

Figure: 5(a)

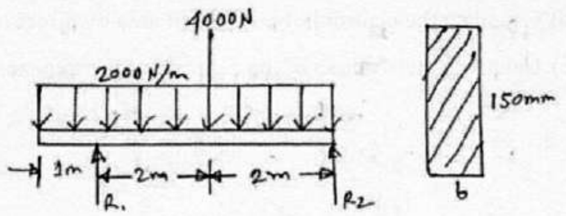

Figure: 5(b)

Q 6. (a) A solid shaft is loaded as shown in figure 6(a) below. Using $G = 83\ GPa$, determine the required diameter of the shaft if the shearing stress is limited to $50\ MPa$ and the angle of rotation at the free end is not to exceed 4°. (15)

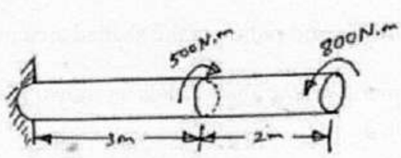

Figure: 6(a)

(b) A simply supported beam carries a couple M applied as shown in figure 6(b). Determine the equation of the elastic curve and the deflection at the point of application of the couple. (15)

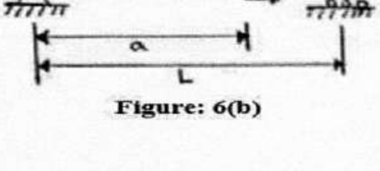

Figure: 6(b)

Khulna University of Engineering & Technology
B.Sc. Engineering 2nd Year 1st Term Examination, 2020
Department of Materials Science and Engineering

Hum 2127
(Accounting and Economics)

Time: 1.5 hours Full Marks: 120

N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY TWO** questions from this section in **answer script A**)

Q 1. (a) Define accounting. How can you use accounting knowledge as an engineer? (10)

(b) Define transaction. State the characteristics of transaction. (12)

(c) What is accounting equation? Mention the real elements of accounting equation. (08)

Q 2. Mr. Hafiz started a business with cash Tk. 300,000 and furniture worth Tk. 50,000 on 1st January 2021. Other transactions are as follows: (30)

January-2 cash deposited into bank Tk. 80,000.
January-3 office supplies purchased by cheque Tk. 7000.
January-4 Advertisement bill received but not yet paid Tk. 4000.
January-5 Hired an office assistant for the month of salary Tk. 10,000.
January-6 Advertisement bill paid by cheque transaction January-4.
January-7 Insurance premium paid in cash Tk. 2000.
January-8 Services provided and billed customers Tk. 70,000.

Required: a) Prepare journal and b) Prepare Tabular summary. The column headings should be as follows: cash + Accounts receivable + supplies +bank +furniture = Accounts payable + Hafiz capital.

Q 3. (a) What are the errors not detected by a trial balance? Explain. (10)

(b) The following accounts are taken from the ledger of M. Rahman trading company at December 31, 2021. (20)

Name of Accounts	**Tk.**
Cash at bank	43,400
Sales	48,500
Sales return	2,800
Depreciation	700
Allowance for bad debit	31,000
Accumulated depreciation	19,500
Advance paid salaries	31,100
Interest received	67,000
Rent receivable	27,550
Motor car	41,500
Profit on sales	11,250
Bills payable	20,250
Income tax expense	26,700
Loss on fire	2,650
Goodwill	21,100

Section B

(Answer **ANY TWO** questions from this section in **answer script B**)

Q 4.	(a) Definition of micro economics and macro economics.	(10)
	(b) Why is the slope of demand negative?	(10)
	(c) What do you mean by cross elasticity? How can you measure cross elasticity?	(10)
Q 5.	(a) What do you mean by indifference curve? Mention the characteristics of indifference curve.	(08)
	(b) Why the sort-run average cost curve is "U" shaped.	(10)
	(c) Explain producer's equilibrium by cost minimization on the basis of fixed production with the help of graph.	(12)
Q 6.	(a) "Monopoly is inefficient in comparison to perfect competition"-Do you think so?	(15)
	(b) Describe the major problems of measuring national income in developing countries like Bangladesh?	(15)